Tutorial Sheet: Power Flow Analysis

IIT Delhi - Abu Dhabi

Course Code: AENL226 - Power Electronics and Power Systems

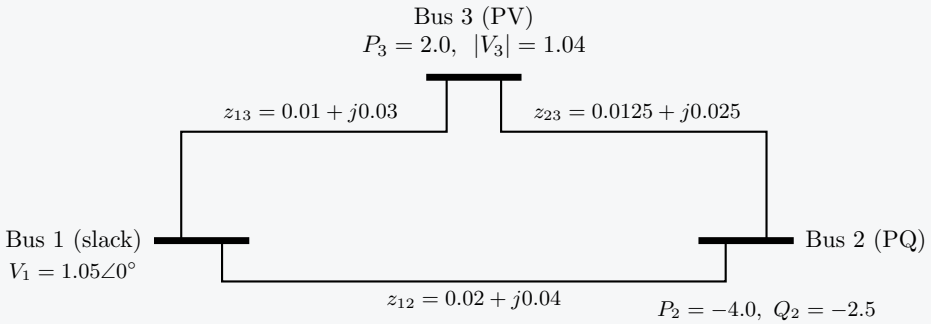
Gauss–Seidel and Newton–Raphson

Unless otherwise stated, all quantities are in per unit. Angles in the Newton–Raphson method are in radians; angles reported with a degree sign are in degrees.

Question 1: Gauss–Seidel Method

Question 1

Consider the three-bus system shown below. Bus 1 is the slack bus, Bus 2 is a PQ bus and Bus 3 is a PV bus. The reactive-power limit at Bus 3 is $1.0 \le Q_3 \le 1.5$.



Using the Gauss–Seidel method with the initial values $V_2^{(0)} = 1.0\angle 0^\circ$ and $V_3^{(0)} = 1.04\angle 0^\circ$, perform the first two iterations.

(a) Form the bus-admittance matrix Y_{bus}.

(b) Determine the updated values of $|V_2|$, δ_2, Q_3 and δ_3 after the first iteration.

(c) Check the reactive-power limits at Bus 3 and state whether it remains a PV bus after the first iteration.

(d) Perform the second iteration and repeat the reactive-power limit check.

(a) Bus-admittance matrix

The series-line admittances are $y_{ij} = 1/z_{ij}$:

$$y_{12} = \frac{1}{0.02 + j0.04} = \frac{0.02 - j0.04}{0.02^2 + 0.04^2} = 10 - j20,$$
$$y_{13} = \frac{1}{0.01 + j0.03} = \frac{0.01 - j0.03}{0.01^2 + 0.03^2} = 10 - j30,$$
$$y_{23} = \frac{1}{0.0125 + j0.025} = \frac{0.0125 - j0.025}{0.0125^2 + 0.025^2} = 16 - j32.$$

For a series branch between buses i and j, its admittance is added to both diagonal entries and its negative is placed in the two off-diagonal entries:

$$\begin{aligned} Y_{11} &= y_{12} + y_{13} = 20 - j50, \quad Y_{12} = Y_{21} = -y_{12} = -10 + j20, \\ Y_{22} &= y_{12} + y_{23} = 26 - j52, \quad Y_{13} = Y_{31} = -y_{13} = -10 + j30, \\ Y_{33} &= y_{13} + y_{23} = 26 - j62, \quad Y_{23} = Y_{32} = -y_{23} = -16 + j32. \end{aligned}$$

$$Y_{\text{bus}} = \begin{bmatrix} 20 - j50 & -10 + j20 & -10 + j30 \\ -10 + j20 & 26 - j52 & -16 + j32 \\ -10 + j30 & -16 + j32 & 26 - j62 \end{bmatrix}$$

(b) First Gauss–Seidel iteration

Initial values: $V_1 = 1.05\angle 0^\circ$ (slack, fixed), $V_2^{(0)} = 1.0\angle 0^\circ$, $V_3^{(0)} = 1.04\angle 0^\circ$.

The Gauss–Seidel update for bus i is

$$V_i^{(k+1)} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{\left(V_i^{(k)}\right)^*} - \sum_{k' \neq i} Y_{ik'} V_{k'} \right],$$

where each $V_{k'}$ is the most recent available value.

Bus 2 (PQ). With $P_2 - jQ_2 = -4 - j(-2.5) = -4 + j2.5$:

$$\begin{aligned} V_2^{(1)} &= \frac{1}{26 - j52} \left[\frac{-4 + j2.5}{(1\angle 0^\circ)^*} - (-10 + j20)(1.05) - (-16 + j32)(1.04) \right] \\ &= \frac{1}{26 - j52} \left[(-4 + j2.5) + (10.5 - j21) + (16.64 - j33.28) \right] \\ &= \frac{23.14 - j51.78}{26 - j52} = 0.9746 - j0.0423 = 0.9755\angle -2.4856^\circ \end{aligned}$$

$$|V_2^{(1)}| = 0.9755, \qquad \delta_2^{(1)} = -2.4856^\circ$$

Bus 3 (PV): reactive power. Q_3 is not specified, so it is first calculated from the latest voltages ($V_2^{(1)}$ is already available):

$$Q_3^{(1)} = -\operatorname{Im}\left\{ \left(V_3^{(0)}\right)^* \left(Y_{31} V_1 + Y_{32} V_2^{(1)} + Y_{33} V_3^{(0)}\right) \right\}$$

The three products are

$$\begin{aligned} Y_{31} V_1 &= (-10 + j30)(1.05) = -10.5000 + j31.5000 \\ Y_{32} V_2^{(1)} &= (-16 + j32)(0.974615 - j0.042308) = -14.2400 + j31.8646 \\ Y_{33} V_3^{(0)} &= (26 - j62)(1.04) = 27.0400 - j64.4800 \end{aligned}$$

Sum $= 2.3000 - j1.1154$. Multiplying by $(V_3^{(0)})^* = 1.04$ gives $2.3920 - j1.1600$, so

$$Q_3^{(1)} = -(-1.1600) = 1.1600 \quad \text{(within limits)}$$

Bus 3 (PV): angle. Use $Q_3^{(1)}$ to compute a provisional voltage:

$$\begin{aligned} V_3' &= \frac{1}{Y_{33}}\left[\frac{P_3 - jQ_3^{(1)}}{\left(V_3^{(0)}\right)^*} - Y_{31}V_1 - Y_{32}V_2^{(1)}\right] \\ &= \frac{1}{26 - j62}\left[\frac{2.0 - j1.16}{1.04} + (10.5 - j31.5) + (14.24 - j31.8646)\right] \\ &= \frac{26.6631 - j64.4800}{26 - j62} \approx 1.0378 - j0.0052 = 1.0378\angle -0.2854^\circ \end{aligned}$$

Since Bus 3 is a PV bus, its specified magnitude is retained and only the angle is kept:

$$|V_3^{(1)}| = 1.04, \qquad \delta_3^{(1)} = -0.2854^\circ, \qquad V_3^{(1)} = 1.04\angle -0.2854^\circ = 1.0400 - j0.0052$$

(c) Reactive-power limit check

$$1.0 \le Q_3^{(1)} = 1.1600 \le 1.5$$

The limit is not violated, so **Bus 3 remains a PV bus** after the first Gauss–Seidel iteration.

(d) Second Gauss–Seidel iteration

Start from $V_1 = 1.05\angle 0^\circ$, $V_2^{(1)} = 0.974615 - j0.042308$ and $V_3^{(1)} = 1.039987 - j0.005181 = 1.04\angle -0.2854^\circ$.

Bus 2 (PQ). $V_3^{(1)}$ is now used in place of the initial value:

$$\begin{aligned} \frac{-4 + j2.5}{\left(V_2^{(1)}\right)^*} &= \frac{-4 + j2.5}{0.974615 + j0.042308} = -3.9853 + j2.7381 \\ -Y_{21}V_1 &= 10.5000 - j21.0000 \\ -Y_{23}V_3^{(1)} &= -(-16 + j32)(1.039987 - j0.005181) = 16.4740 - j33.3625 \\ V_2^{(2)} &= \frac{22.9887 - j51.6244}{26 - j52} = 0.9711 - j0.0434 = 0.9720\angle -2.5613^\circ \end{aligned}$$

$$|V_2^{(2)}| = 0.9720, \qquad \delta_2^{(2)} = -2.5613^\circ$$

Bus 3 (PV): reactive power, using $V_2^{(2)}$ and $V_3^{(1)}$:

$$\begin{aligned} Y_{31}V_1 &= -10.5000 + j31.5000 \\ Y_{32}V_2^{(2)} &= (-16 + j32)(0.971057 - j0.043438) = -14.1469 + j31.7688 \\ Y_{33}V_3^{(1)} &= (26 - j62)(1.039987 - j0.005181) = 26.7184 - j64.6139 \end{aligned}$$

Sum $= 2.0716 - j1.3451$. Multiplying by $\left(V_3^{(1)}\right)^* = 1.039987 + j0.005181$ gives $2.1614 - j1.3881$, so

$$Q_3^{(2)} = 1.3881$$

Bus 3 (PV): angle.

$$\begin{aligned} V_3' &= \frac{1}{26 - j62}\left[\frac{2.0 - j1.3881}{1.039987 + j0.005181} + (10.5 - j31.5) + (14.1469 - j31.7688)\right] \\ &= \frac{1}{26 - j62}\left[(1.9164 - j1.3443) + (10.5 - j31.5) + (14.1469 - j31.7688)\right] \\ &= \frac{26.5633 - j64.6131}{26 - j62} \approx 1.0391 - j0.0073 = 1.0391\angle -0.4028^\circ \end{aligned}$$

Retaining the specified magnitude:

$$|V_3^{(2)}| = 1.04, \qquad \delta_3^{(2)} = -0.4028^\circ, \qquad V_3^{(2)} = 1.04\angle -0.4028^\circ = 1.0400 - j0.0073$$

Limit check. $1.0 \le Q_3^{(2)} = 1.3881 \le 1.5 \;\Rightarrow$ **Bus 3 remains a PV bus**.

Convergence. The largest voltage change in iteration 2 is $|V_2^{(2)} - V_2^{(1)}| = 0.0037$ (bus 3: 0.0021), still well above a typical tolerance of 10^{-4}. Gauss–Seidel needs several more iterations to converge.

Question 2: Newton–Raphson Method

Question 2

Repeat the first two iterations using the Newton–Raphson method. Start from $|V_2| = 1.0$, $\delta_2 = 0^\circ$ and $\delta_3 = 0^\circ$, with $|V_3| = 1.04$.

(a) Form Y_{bus}, calculate the initial bus powers, and obtain the power-mismatch vector.

(b) Form and evaluate the Newton–Raphson Jacobian matrix.

(c) Compute the first update vector and obtain the updated state variables.

(d) Calculate Q_3 using the updated voltages and check its reactive-power limits. State the Bus 3 type for the next iteration. If a bus-type conversion is required, identify the corresponding changes in the Newton–Raphson formulation.

(e) Perform the second iteration and repeat the reactive-power limit check.

Show the principal intermediate quantities, including Y_{bus}, calculated powers, power mismatches, Jacobian elements, update vector and reactive-power limit check. Report voltage angles in degrees.

(a) Y_{bus}, initial bus powers and mismatches

Y_{bus} is as in Question 1. Writing $Y_{ij} = G_{ij} + jB_{ij}$:

$$G = \begin{bmatrix} 20 & -10 & -10 \\ -10 & 26 & -16 \\ -10 & -16 & 26 \end{bmatrix}, \qquad B = \begin{bmatrix} -50 & 20 & 30 \\ 20 & -52 & 32 \\ 30 & 32 & -62 \end{bmatrix}.$$

With $\delta_{ij} = \delta_i - \delta_j$, the calculated powers are

$$P_i^{\text{calc}} = |V_i| \sum_{j=1}^{n} |V_j| \big(G_{ij} \cos \delta_{ij} + B_{ij} \sin \delta_{ij}\big),$$

$$Q_i^{\text{calc}} = |V_i| \sum_{j=1}^{n} |V_j| \big(G_{ij} \sin \delta_{ij} - B_{ij} \cos \delta_{ij}\big).$$

At the initial point $|V_1| = 1.05$, $|V_2| = 1.0$, $|V_3| = 1.04$ and all $\delta_{ij} = 0$, so $\cos \delta_{ij} = 1$, $\sin \delta_{ij} = 0$:

$$P_2^{\text{calc}} = 1.0\,[1.05(-10) + 1.0(26) + 1.04(-16)] = -10.5000 + 26.0000 - 16.6400 = -1.1400$$
$$P_3^{\text{calc}} = 1.04\,[1.05(-10) + 1.0(-16) + 1.04(26)] = 1.04\,[-10.5000 - 16.0000 + 27.0400] = 0.5616$$
$$Q_2^{\text{calc}} = 1.0\,[-1.05(20) - 1.0(-52) - 1.04(32)] = -21.0000 + 52.0000 - 33.2800 = -2.2800$$
$$Q_3^{\text{calc}} = 1.04\,[-1.05(30) - 1.0(32) - 1.04(-62)] = 1.04\,[-31.5000 - 32.0000 + 64.4800] = 1.0192$$

(Q_3^{calc} is not a mismatch quantity, since Bus 3 is PV, but it is needed in the Jacobian.)

Mismatches $\Delta P_i = P_i^{sp} - P_i^{\text{calc}}$, $\Delta Q_i = Q_i^{sp} - Q_i^{\text{calc}}$:

$$\Delta P_2 = -4 - (-1.1400) = -2.8600$$
$$\Delta P_3 = 2 - 0.5616 = 1.4384$$
$$\Delta Q_2 = -2.5 - (-2.2800) = -0.2200$$

$$\Delta y^{(0)} = \begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta Q_2 \end{bmatrix} = \begin{bmatrix} -2.8600 \\ 1.4384 \\ -0.2200 \end{bmatrix}$$

Derivation of the power equations and Jacobian terms

Start from the injected current and complex power at bus i:

$$I_i = \sum_{j=1}^{n} Y_{ij} V_j, \qquad S_i = P_i^{\text{calc}} + jQ_i^{\text{calc}} = V_i I_i^*.$$

With $V_i = |V_i| e^{j\delta_i}$ and $Y_{ij} = G_{ij} + jB_{ij}$,

$$S_i = |V_i| e^{j\delta_i} \sum_{j=1}^{n} (G_{ij} - jB_{ij}) |V_j| e^{-j\delta_j} = |V_i| \sum_{j=1}^{n} |V_j| (G_{ij} - jB_{ij}) e^{j\delta_{ij}}.$$

Expanding $e^{j\delta_{ij}} = \cos \delta_{ij} + j \sin \delta_{ij}$:

$$(G_{ij} - jB_{ij}) e^{j\delta_{ij}} = (G_{ij} \cos \delta_{ij} + B_{ij} \sin \delta_{ij}) + j(G_{ij} \sin \delta_{ij} - B_{ij} \cos \delta_{ij}),$$

and equating real and imaginary parts gives the expressions for P_i^{calc} and Q_i^{calc} above.
For compact differentiation define

$$A_{ij} = G_{ij} \cos \delta_{ij} + B_{ij} \sin \delta_{ij}, \qquad C_{ij} = G_{ij} \sin \delta_{ij} - B_{ij} \cos \delta_{ij},$$

so $P_i^{\text{calc}} = |V_i| \sum_j |V_j| A_{ij}$ and $Q_i^{\text{calc}} = |V_i| \sum_j |V_j| C_{ij}$, with $\partial A_{ij}/\partial \delta_{ij} = -C_{ij}$ and $\partial C_{ij}/\partial \delta_{ij} = A_{ij}$ (angles in radians). Note $A_{ii} = G_{ii}$ and $C_{ii} = -B_{ii}$.
Off-diagonal ($i \neq j$). Since $\partial \delta_{ij}/\partial \delta_j = -1$:

$$\frac{\partial P_i^{\text{calc}}}{\partial \delta_j} = |V_i||V_j| C_{ij}, \quad \frac{\partial Q_i^{\text{calc}}}{\partial \delta_j} = -|V_i||V_j| A_{ij}, \quad \frac{\partial P_i^{\text{calc}}}{\partial |V_j|} = |V_i| A_{ij}, \quad \frac{\partial Q_i^{\text{calc}}}{\partial |V_j|} = |V_i| C_{ij}.$$

Diagonal, angle. Changing δ_i changes every δ_{ij} ($j \neq i$), while $\delta_{ii} = 0$:

$$\frac{\partial P_i^{\text{calc}}}{\partial \delta_i} = -|V_i| \sum_{j \neq i} |V_j| C_{ij} = -\big(Q_i^{\text{calc}} - |V_i|^2 C_{ii}\big) = -Q_i^{\text{calc}} - B_{ii} |V_i|^2,$$

$$\frac{\partial Q_i^{\text{calc}}}{\partial \delta_i} = |V_i| \sum_{j \neq i} |V_j| A_{ij} = P_i^{\text{calc}} - G_{ii}|V_i|^2.$$

Diagonal, magnitude. $|V_i|$ appears both as the leading factor and in the $j = i$ term (product rule):

$$\frac{\partial P_i^{\text{calc}}}{\partial |V_i|} = 2|V_i|A_{ii} + \sum_{j \neq i} |V_j| A_{ij} = \frac{P_i^{\text{calc}}}{|V_i|} + G_{ii}|V_i|, \qquad \frac{\partial Q_i^{\text{calc}}}{\partial |V_i|} = \frac{Q_i^{\text{calc}}}{|V_i|} - B_{ii}|V_i|.$$

(b) Jacobian

Unknowns $[\delta_2,\ \delta_3,\ |V_2|]^T$; mismatches $[\Delta P_2,\ \Delta P_3,\ \Delta Q_2]^T$. At the initial point ($\delta_{ij} = 0$):

$$\frac{\partial P_2}{\partial \delta_2} = -Q_2^{\text{calc}} - B_{22}|V_2|^2 = 2.2800 + 52 = 54.2800$$
$$\frac{\partial P_2}{\partial \delta_3} = |V_2||V_3|(G_{23}\sin\delta_{23} - B_{23}\cos\delta_{23}) = (1)(1.04)(-32) = -33.2800$$
$$\frac{\partial P_2}{\partial |V_2|} = \frac{P_2^{\text{calc}}}{|V_2|} + G_{22}|V_2| = -1.1400 + 26 = 24.8600$$
$$\frac{\partial P_3}{\partial \delta_2} = |V_3||V_2|(G_{32}\sin\delta_{32} - B_{32}\cos\delta_{32}) = (1.04)(1)(-32) = -33.2800$$
$$\frac{\partial P_3}{\partial \delta_3} = -Q_3^{\text{calc}} - B_{33}|V_3|^2 = -1.0192 + 62(1.04)^2 = -1.0192 + 67.0592 = 66.0400$$
$$\frac{\partial P_3}{\partial |V_2|} = |V_3|(G_{32}\cos\delta_{32} + B_{32}\sin\delta_{32}) = 1.04(-16) = -16.6400$$
$$\frac{\partial Q_2}{\partial \delta_2} = P_2^{\text{calc}} - G_{22}|V_2|^2 = -1.1400 - 26 = -27.1400$$
$$\frac{\partial Q_2}{\partial \delta_3} = -|V_2||V_3|(G_{23}\cos\delta_{23} + B_{23}\sin\delta_{23}) = -(1)(1.04)(-16) = 16.6400$$
$$\frac{\partial Q_2}{\partial |V_2|} = \frac{Q_2^{\text{calc}}}{|V_2|} - B_{22}|V_2| = -2.2800 + 52 = 49.7200$$

$$J^{(0)} = \begin{bmatrix} 54.2800 & -33.2800 & 24.8600 \\ -33.2800 & 66.0400 & -16.6400 \\ -27.1400 & 16.6400 & 49.7200 \end{bmatrix}$$

(c) Update vector and updated state

Solve $J^{(0)}\Delta x^{(0)} = \Delta y^{(0)}$:

$$\begin{bmatrix} 54.2800 & -33.2800 & 24.8600 \\ -33.2800 & 66.0400 & -16.6400 \\ -27.1400 & 16.6400 & 49.7200 \end{bmatrix} \begin{bmatrix} \Delta\delta_2 \\ \Delta\delta_3 \\ \Delta|V_2| \end{bmatrix} = \begin{bmatrix} -2.8600 \\ 1.4384 \\ -0.2200 \end{bmatrix} \quad \Longrightarrow \quad \Delta x^{(0)} = \begin{bmatrix} -0.045263 \\ -0.007718 \\ -0.026549 \end{bmatrix}$$

The angle corrections are in radians, because the Jacobian was obtained by differentiating with respect to radian-valued angles:

$$\delta_2^{(1)} = 0 + (-0.045263)\tfrac{180^\circ}{\pi} = -2.5934^\circ$$
$$\delta_3^{(1)} = 0 + (-0.007718)\tfrac{180^\circ}{\pi} = -0.4422^\circ$$
$$|V_2|^{(1)} = 1.0 - 0.026549 = 0.9735$$

$V_2^{(1)} = 0.9735\angle -2.5934^\circ$, $V_3^{(1)} = 1.04\angle -0.4422^\circ$ (magnitude held, PV bus).

(d) Reactive-power limit check at Bus 3

$$Q_3^{(1)} = |V_3| \sum_{j=1}^{3} |V_j| \left(G_{3j} \sin \delta_{3j} - B_{3j} \cos \delta_{3j}\right)$$

with $\delta_{31} = \delta_3 - \delta_1 = -0.4422^\circ$ and $\delta_{32} = \delta_3 - \delta_2 = -0.4422^\circ + 2.5934^\circ = 2.1511^\circ$:

$$\begin{aligned} Q_3^{(1)} &= 1.04\Big[1.05\{(-10)\sin(-0.4422^\circ) - 30\cos(-0.4422^\circ)\} \\ &\qquad + 0.9735\{(-16)\sin 2.1511^\circ - 32\cos 2.1511^\circ\} + 1.04\{26\sin 0^\circ + 62\cos 0^\circ\}\Big] \\ &= 1.04\,[-31.4180 - 31.7131 + 64.4800] = 1.4028 \end{aligned}$$

$1.0 \le Q_3^{(1)} = 1.4028 \le 1.5$: the limit is satisfied, so **Bus 3 remains a PV bus** for the next iteration. No change to the Newton–Raphson formulation is needed.

Had a limit been violated, Q_3 would be fixed at the violated limit ($Q_{3,\min}$ or $Q_{3,\max}$) and Bus 3 treated as a PQ bus: $|V_3|$ would become an additional unknown (starting from 1.04), $\Delta Q_3 = Q_3^{sp} - Q_3^{\text{calc}}$ would become an additional mismatch equation, and the Jacobian would grow from 3×3 to 4×4.

(e) Second Newton–Raphson iteration

Bus 3 is still a PV bus, so the unknowns and mismatches are the same as before. State after iteration 1 (angles in radians):

$$\delta_2^{(1)} = -0.045263, \quad \delta_3^{(1)} = -0.007718, \quad |V_2|^{(1)} = 0.973451, \quad |V_1| = 1.05,\ |V_3| = 1.04.$$

Angle differences: $\delta_{21} = -2.5934^\circ$, $\delta_{23} = \delta_2 - \delta_3 = -2.1511^\circ$, $\delta_{31} = -0.4422^\circ$, $\delta_{32} = 2.1511^\circ$.

Calculated powers. Listing the $j = 1, 2, 3$ terms of each sum:

$$P_2^{\text{calc}} = -11.1357 + 24.6378 - 17.4028 = -3.9008$$
$$P_3^{\text{calc}} = -11.1725 - 14.9708 + 28.1216 = 1.9783$$
$$Q_2^{\text{calc}} = -19.9591 + 49.2756 - 31.7656 = -2.4491$$
$$Q_3^{\text{calc}} = -32.6747 - 32.9816 + 67.0592 = 1.4028$$

Mismatches.

$$\Delta P_2 = -4 - (-3.9008) = -0.0992$$
$$\Delta P_3 = 2 - 1.9783 = 0.0217$$
$$\Delta Q_2 = -2.5 - (-2.4491) = -0.0509$$

The mismatches have fallen from about 2.9 to about 0.1 after one iteration.

Jacobian at the new point.

$$\begin{aligned}
\frac{\partial P_2}{\partial \delta_2} &= -Q_2^{\text{calc}} - B_{22}|V_2|^2 = 2.4491 + 52(0.973451)^2 = 51.7247 \\
\frac{\partial P_2}{\partial \delta_3} &= |V_2||V_3|(G_{23}\sin\delta_{23} - B_{23}\cos\delta_{23}) \\
&= (0.973451)(1.04)\big[-16\sin(-2.1511^\circ) - 32\cos(-2.1511^\circ)\big] = -31.7656 \\
\frac{\partial P_2}{\partial |V_2|} &= \frac{P_2^{\text{calc}}}{|V_2|} + G_{22}|V_2| = \frac{-3.9008}{0.973451} + 26(0.973451) = 21.3026 \\
\frac{\partial P_3}{\partial \delta_2} &= |V_3||V_2|(G_{32}\sin\delta_{32} - B_{32}\cos\delta_{32}) \\
&= (1.04)(0.973451)\big[-16\sin 2.1511^\circ - 32\cos 2.1511^\circ\big] = -32.9816 \\
\frac{\partial P_3}{\partial \delta_3} &= -Q_3^{\text{calc}} - B_{33}|V_3|^2 = -1.4028 + 62(1.04)^2 = 65.6564 \\
\frac{\partial P_3}{\partial |V_2|} &= |V_3|(G_{32}\cos\delta_{32} + B_{32}\sin\delta_{32}) \\
&= 1.04\big[-16\cos 2.1511^\circ + 32\sin 2.1511^\circ\big] = -15.3791 \\
\frac{\partial Q_2}{\partial \delta_2} &= P_2^{\text{calc}} - G_{22}|V_2|^2 = -3.9008 - 26(0.973451)^2 = -28.5386 \\
\frac{\partial Q_2}{\partial \delta_3} &= -|V_2||V_3|(G_{23}\cos\delta_{23} + B_{23}\sin\delta_{23}) \\
&= -(0.973451)(1.04)\big[-16\cos(-2.1511^\circ) + 32\sin(-2.1511^\circ)\big] = 17.4028 \\
\frac{\partial Q_2}{\partial |V_2|} &= \frac{Q_2^{\text{calc}}}{|V_2|} - B_{22}|V_2| = \frac{-2.4491}{0.973451} + 52(0.973451) = 48.1036
\end{aligned}$$

$$J^{(1)} = \begin{bmatrix} 51.7247 & -31.7656 & 21.3026 \\ -32.9816 & 65.6564 & -15.3791 \\ -28.5386 & 17.4028 & 48.1036 \end{bmatrix}, \qquad \Delta y^{(1)} = \begin{bmatrix} -0.0992 \\ 0.0217 \\ -0.0509 \end{bmatrix}$$

Update. Solving $J^{(1)}\Delta x^{(1)} = \Delta y^{(1)}$:

$$\Delta x^{(1)} = \begin{bmatrix} \Delta\delta_2 \\ \Delta\delta_3 \\ \Delta|V_2| \end{bmatrix} = \begin{bmatrix} -0.001795 \\ -0.000985 \\ -0.001767 \end{bmatrix}$$

$$\begin{aligned}
\delta_2^{(2)} &= -0.045263 - 0.001795 = -0.047058 \text{ rad} = -2.6962^\circ \\
\delta_3^{(2)} &= -0.007718 - 0.000985 = -0.008703 \text{ rad} = -0.4987^\circ \\
|V_2|^{(2)} &= 0.973451 - 0.001767 = 0.9717
\end{aligned}$$

$$V_2^{(2)} = 0.9717\angle -2.6962^\circ, \qquad V_3^{(2)} = 1.04\angle -0.4987^\circ$$

Limit check. With the updated voltages, $Q_3^{(2)} = 1.4616$. Since $1.0 \le 1.4616 \le 1.5$, **Bus 3 remains a PV bus**.

Convergence. At this new point the largest mismatch is about 2.2×10^{-4} (it was 2.86 at the start and 0.099 after iteration 1). One more iteration brings it well below 10^{-4}, showing the quadratic convergence typical of Newton–Raphson.

Comparison of the two methods

Method	**Iteration**	$\lvert V_2\rvert$	δ_2	δ_3	Q_3
Gauss–Seidel	1	0.9755	-2.4856°	-0.2854°	1.1600
	2	0.9720	-2.5613°	-0.4028°	1.3881
Newton–Raphson	1	0.9735	-2.5934°	-0.4422°	1.4028
	2	0.9717	-2.6962°	-0.4987°	1.4616
Converged solution	–	0.9717	-2.6965°	-0.4988°	1.4618

After two iterations Newton–Raphson agrees with the converged solution to about four decimal places, while Gauss–Seidel is still noticeably off (e.g. δ_3 is -0.4028° against -0.4988°). Each Newton–Raphson iteration costs more (Jacobian formation), but far fewer iterations are needed.